

Model Predictor Testing and Optimization

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

17 March 2026

Contents

Model I 1

Functions Used: modfunctions_notes.R + modelfunctions.R

Model I

Table 1: LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19714.49	45	NA	NA	0.1662	0.5329	0.4302	NA	NA	31281.11	31622.16

\begin{table}

\caption{LN_SPPSF model coefficients (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.0550	0.0137	296.4117	0.0000	NA	NA	NA	NA
parametric	arterial	-0.0185	0.0146	-1.2639	0.2063	NA	NA	NA	NA
parametric	single_family1story	0.0433	0.0116	3.7189	0.0002	NA	NA	NA	NA
parametric	half_duplex1story	0.0602	0.0484	1.2446	0.2133	NA	NA	NA	NA
parametric	half_duplex2story	0.0318	0.0464	0.6840	0.4940	NA	NA	NA	NA
parametric	smallest	0.1291	0.0152	8.4868	0.0000	NA	NA	NA	NA
parametric	small	0.0479	0.0111	4.2977	0.0000	NA	NA	NA	NA
parametric	large	-0.0040	0.0126	-0.3181	0.7504	NA	NA	NA	NA
parametric	largest	0.0178	0.0230	0.7726	0.4398	NA	NA	NA	NA
parametric	two_plus_baths	0.1448	0.0274	5.2770	0.0000	NA	NA	NA	NA
parametric	powder_room	0.0659	0.0159	4.1324	0.0000	NA	NA	NA	NA
parametric	good	0.3567	0.0294	12.1144	0.0000	NA	NA	NA	NA
parametric	fair	-0.2076	0.0095	-21.8315	0.0000	NA	NA	NA	NA
parametric	poor	-0.3609	0.0217	-16.6150	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.3365	0.0453	-7.4290	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.4634	0.0284	16.3392	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0380	0.0100	-3.8082	0.0001	NA	NA	NA	NA
parametric	crawl	-0.0138	0.0265	-0.5211	0.6023	NA	NA	NA	NA
parametric	slab	-0.0522	0.0313	-1.6639	0.0962	NA	NA	NA	NA
parametric	fireplace	0.1045	0.0097	10.7497	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0769	0.0150	5.1129	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2946	0.1437	-2.0506	0.0403	NA	NA	NA	NA
parametric	wall	-0.1005	0.0501	-2.0063	0.0448	NA	NA	NA	NA
parametric	central_air	0.0921	0.0108	8.5470	0.0000	NA	NA	NA	NA
smooth	s(sresb_yearbuilt)	NA	NA	55.4700	0.0000	NA	NA	7.6054	9
smooth	s(ln_garagespaces)	NA	NA	39.8021	0.0000	NA	NA	1.9035	3
smooth	s(months)	NA	NA	66.1422	0.0000	NA	NA	7.9988	9

\end{table}

Table 2: LN_SPPSF model fit statistics1

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19712.03	59	NA	NA	0.1665	0.5328	0.4302	NA	NA	31281.91	31658.11

\begin{table}

\caption{LN_SPPSF model coefficients1 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.0658	0.0155	261.9751	0.0000	NA	NA	NA	NA
parametric	arterial	-0.0183	0.0146	-1.2500	0.2113	NA	NA	NA	NA
parametric	single_family1story	0.0419	0.0117	3.5855	0.0003	NA	NA	NA	NA
parametric	half_duplex1story	0.0553	0.0485	1.1412	0.2538	NA	NA	NA	NA
parametric	half_duplex2story	0.0262	0.0466	0.5619	0.5742	NA	NA	NA	NA
parametric	smallest	0.0936	0.0275	3.3975	0.0007	NA	NA	NA	NA
parametric	small	0.0321	0.0149	2.1592	0.0308	NA	NA	NA	NA
parametric	large	0.0110	0.0163	0.6770	0.4984	NA	NA	NA	NA
parametric	largest	0.0432	0.0293	1.4744	0.1404	NA	NA	NA	NA
parametric	two_plus_baths	0.0807	0.0599	1.3471	0.1780	NA	NA	NA	NA
parametric	powder_room	0.0291	0.0311	0.9365	0.3490	NA	NA	NA	NA
parametric	good	0.3561	0.0294	12.0942	0.0000	NA	NA	NA	NA
parametric	fair	-0.2079	0.0095	-21.8597	0.0000	NA	NA	NA	NA
parametric	poor	-0.3612	0.0217	-16.6308	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.3374	0.0453	-7.4495	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.4616	0.0294	15.6931	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0396	0.0100	-3.9500	0.0001	NA	NA	NA	NA
parametric	crawl	-0.0148	0.0265	-0.5598	0.5756	NA	NA	NA	NA
parametric	slab	-0.0525	0.0313	-1.6762	0.0937	NA	NA	NA	NA
parametric	fireplace	0.1061	0.0098	10.8547	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0766	0.0150	5.0912	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2882	0.1437	-2.0061	0.0449	NA	NA	NA	NA
parametric	wall	-0.1032	0.0501	-2.0580	0.0396	NA	NA	NA	NA
parametric	central_air	0.0923	0.0108	8.5638	0.0000	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	0.3690	0.0405	NA	NA	1.0507	9
smooth	s(totalbath)	NA	NA	0.7473	0.0341	NA	NA	1.4202	5
smooth	s(sresb_yearbuilt)	NA	NA	55.3197	0.0000	NA	NA	7.6079	9
smooth	s(ln_garagespaces)	NA	NA	39.7589	0.0000	NA	NA	1.9003	3
smooth	s(months)	NA	NA	66.2831	0.0000	NA	NA	7.9945	9

\end{table}

Table 3: LN_SPPSF model fit statistics2

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19665.55	101	NA	NA	0.2626	0.5007	0.4053	NA	NA	28893.59	29582.4

\begin{table}

\caption{LN_SPPSF model coefficients2 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.0956	0.0156	263.0623	0.0000	NA	NA	NA	NA
parametric	arterial	0.0058	0.0139	0.4130	0.6796	NA	NA	NA	NA
parametric	single_family1story	0.0430	0.0111	3.8889	0.0001	NA	NA	NA	NA
parametric	half_duplex1story	-0.0123	0.0461	-0.2667	0.7897	NA	NA	NA	NA
parametric	half_duplex2story	-0.0377	0.0442	-0.8531	0.3936	NA	NA	NA	NA
parametric	smallest	0.0678	0.0297	2.2798	0.0226	NA	NA	NA	NA
parametric	small	0.0243	0.0153	1.5840	0.1132	NA	NA	NA	NA
parametric	large	0.0016	0.0169	0.0974	0.9224	NA	NA	NA	NA
parametric	largest	0.0325	0.0303	1.0714	0.2840	NA	NA	NA	NA
parametric	two_plus_baths	0.0449	0.0629	0.7145	0.4749	NA	NA	NA	NA
parametric	powder_room	-0.0117	0.0324	-0.3599	0.7189	NA	NA	NA	NA
parametric	good	0.3055	0.0278	10.9749	0.0000	NA	NA	NA	NA
parametric	fair	-0.1816	0.0090	-20.1715	0.0000	NA	NA	NA	NA
parametric	poor	-0.3223	0.0205	-15.7145	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2860	0.0427	-6.6981	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3871	0.0289	13.3848	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0622	0.0097	-6.4113	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0212	0.0251	-0.8448	0.3983	NA	NA	NA	NA
parametric	slab	-0.0853	0.0297	-2.8755	0.0040	NA	NA	NA	NA
parametric	fireplace	0.0909	0.0095	9.6019	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0713	0.0142	5.0203	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2073	0.1352	-1.5327	0.1254	NA	NA	NA	NA
parametric	wall	-0.1356	0.0474	-2.8613	0.0042	NA	NA	NA	NA
parametric	central_air	0.0722	0.0102	7.0853	0.0000	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	1.4910	0.0002	NA	NA	1.8406	9
smooth	s(totalbath)	NA	NA	1.2556	0.0043	NA	NA	1.6755	5
smooth	s(sresb_yearbuilt)	NA	NA	17.9070	0.0000	NA	NA	6.0246	9
smooth	s(ln_garagespaces)	NA	NA	55.0108	0.0000	NA	NA	1.9881	3
smooth	s(months)	NA	NA	73.8904	0.0000	NA	NA	7.9439	9
smooth	s(dist_henryford_mi)	NA	NA	38.1347	0.0000	NA	NA	6.8401	7
smooth	s(drive_henryford_min)	NA	NA	37.8368	0.0000	NA	NA	6.3004	7
smooth	s(dist_guardian_mi)	NA	NA	49.8633	0.0000	NA	NA	6.3972	7
smooth	s(drive_guardian_min)	NA	NA	79.3687	0.0000	NA	NA	6.4765	7
smooth	s(dist_dequindre_mi)	NA	NA	72.8422	0.0000	NA	NA	6.6377	7
smooth	s(drive_dequindre_min)	NA	NA	64.7843	0.0000	NA	NA	6.3224	7

\end{table}